



East West University
Department of Mathematical and Physical Sciences
Semester: Fall 2025
Course Outline of STA102; Statistics and Probability: Section: 1/2

Course Information: Course Code: STA 102 Course Title: Statistics & Probability Credit Hours: 3	Instructor: Afsana Al Sharmin Senior Lecturer, Department of Mathematical and Physical Sciences, EWU Office: Room # 1006. Email: aas@ewubd.edu
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Days	Course code	Room No.	Class hour	Office hour
Sunday	STA208 (1) STA208 (2)	AB1-401 FUB-404	10.10AM-11.40AM 3.10PM-4.40PM	9.00AM-10.10AM 11.40AM-1.00PM
Monday	STA102(1) STA102(2)	AB1-602 FUB-504	10.10AM-11.40AM 3.10PM-4.40PM	9.00AM-10.10AM 11.40AM-1.00PM
Tuesday	STA208 (1) STA208 (2)	AB1-401 FUB-404	10.10AM-11.40AM 3.10PM-4.40PM	9.00AM-10.10AM 11.40AM-1.00PM
Wednesday	STA102(1) STA102(2)	AB1-602 FUB-504	10.10AM-11.40AM 3.10PM-4.40PM	9.00AM-10.10AM 11.40AM-1.00PM
Thursday				8.30AM-10.00AM

- **Text Books:**
R.E. Walpole, R.H. Myers, S.L. Myers, and K. Ye (2012). Probability & Statistics for Engineers & Scientists, 9th ed. Pearson.
- **Reference Books:**
 - a. Hines, W.W (et.al.), Probability and Statistics in Engineering, 4thed, John Wiley & Sons, Inc,2003 (Latest Version).
 - b. Mendenhall, W., Beaver, R. J. & Beaver, B. M. (2013). Introduction to Probability and Statistics. 14th ed. Duxbury: Thomson Brooks/Cole.
 - c. Devore, J.L., Probability and Statistics for Engineering and the Science, 8th Edition, 2004.
 - d. Sheldon Ross, A first course in Probability, 8th Edition
 - e. Mason And Lind, Statistical Techniques in Business and Economics, 12th Edition

Course material Website: [Corresponding Google Classroom](#)

- **Course Rationale:** We have all come to Statistics in our everyday lives. We talk about average salaries, percentile, ranks, skew, outliers and so on. In fact, many Comparisons that we routinely make involve numbers. We may not be Conscious of it, but we have been using Statistics for a long time. Again, professional working in engineering is often engaged in both the analysis and the design system where system component characteristics are non-deterministic, the understanding and utilization of probability is essential to the description, design and analysis of such system. This course is designed for engineering and applied science. The course introduces the basic concept of Statistics and probability. Topics include basic combinatory, sampling techniques, random variables, probability distributions, random processes, Bayesian inference, hypothesis testing, confidence intervals and linear regression, quality control etc.
- **Teaching Philosophy/Techniques:** Teaching tools include lectures using G-suite, PowerPoint presentation/ multimedia, relevant video tutorial etc. Students are requested to be well prepared in every class based on the previous classes.

- **Assessment Tools:** These include class tests, mid and final exams, assignments, presentations/viva and attendance. Final exam will be arranged according to the EWU academic calendar. Class tests and mid exam dates will be announced in class.

- **Score Distribution:**

Presentation	Viva	Attendance	Assignment	Class Performance/Term Paper	Class Test
5%	5%	5%	5%	5%	15%

Mid Exam	Final Exam
30%	30%

- **Essential Policy Information:**

1. Zero tolerance for cheating in exams.
2. **No makeup for the quizzes.**
3. Makeup exams for mid-terms and final exams may be arranged for appropriate reasons. In this case, the student must apply with supporting documents.
4. Mobile phones should be switched off in the classroom.

- **Course learning outcomes:** After completing the course, students would be able to-

1. Choose appropriate standard sampling techniques for data collection, and present the data with graphs, tables, and numerical summaries.
2. Evaluate the relationship between two or more variables by correlation and regression analysis.
3. Apply basic probability concepts such as conditional probabilities, independence, Bayes Rule to calculate probabilities of events in different engineering problems.
4. Identify and apply discrete and continuous distributions appropriately in engineering problem, such as in quality control and reliability engineering.
5. Demonstrate the concepts of simple stochastic processes in discrete time; namely, Markov chains, Queuing process, and their applications.
6. Calculate and interpret confidence intervals for estimating population parameters.
7. Perform hypothesis test to make decisions regarding population parameters.

Course Syllabus

Introduction: Nature and scope, nature of statistical data, Attributes, and variables, Discrete and continuous variables, graphs, and diagrams; Measure of location: characteristics of an ideal measure, Arithmetic mean, Geometric mean, Harmonic mean, Median, Mode, Quartiles, Deciles, Percentiles; Measure of dispersion: Absolute measure, Relative measure, Range, Standard deviation, Mean deviation, Quartile deviation, Coefficient of dispersion, Coefficient of variation, Skewness, and kurtosis; Regression and correlation: relation between variables, Fitting of regression lines, Simple correlation, multiple correlation, and regression. Theory of probability: Theorem of total, compound and conditional probability, Bayes theorem, Random variable, Discrete and continuous random variables, probability function, expectation of sum and products, concept of Binomial, Poisson and Normal random distribution, Random process, Auto correlation function of a random process, multiple random process, Basic concepts of discrete and continuous probability distributions, Markov process, Queuing process; sampling technique; Test of significance: Test of means, Variance, correlation coefficients and regression coefficients.

Lesson Plan (approximate)

Lecture	Topics to be discussed	Remarks
1-4	Introduction to Statistics: Definition of statistics, scope of statistics, data, population, sample, nature of statistical data, attributes and variables, discrete and continuous variables; Levels of measurement, Methods of data collection. Frequency distribution, Descriptive plots: Bar diagram, Pie-chart, line plot, Histogram, Frequency polygon, Ogive; Stem and leaf plot.	Class Test
5-6	Descriptive statistics: Measure of location- characteristics of an ideal measure, Arithmetic mean, Median, Mode; Measures of dispersion- Range, Quartiles, Deciles and Percentiles.	
7-8	Measures of dispersion- Standard deviation, Mean deviation, Coefficient of variation; Skewness and kurtosis; Box plot, Identifying outliers.	
9-11	Correlation and Regression: Relation between variables, Scatter diagram; Pearson's correlation coefficient, Spearman's rank correlation coefficient; Fitting of regression lines, goodness of fitting model, basic concept of Multiple correlation and regression;	
12-13	Theory of probability: Experiment, Sample space, Event, and mutually exclusive event. Rules of Addition; Rules of multiplication, Conditional probability and independence, Tree diagram, Theorems of total, compound and conditional probability, Bayes theorem.	
	Review	
	Mid Term Exam	
16-18	Random variables: Discrete and continuous random variables, Probability function, Expectation of sum and products, and variance of random variables; Probability distributions: Concepts; Binomial distribution, Poisson distribution;	Class Test
19-20	Probability distributions: Normal distribution (with application of Quality Control); Probability distributions: Exponential distribution with application in reliability; Uniform distribution;	
21-22	Stochastic process: Basic concept of stochastic process; Auto correlation function of a random process, Multiple random process, Markov process;	
23-24	Queuing process: Sampling distribution of mean and proportion,	
25-26	Hypothesis test: Formulation of hypothesis, testing hypothesis using p-value and critical value. Test about Population Mean, Test about Population proportion.	
27-28	Confidence intervals: Confidence intervals for population mean and proportion. Test of significance and confidence interval for Correlation coefficients and Regression coefficients.	
29	Review	
30	Presentation	